

ULTRON

A Fine-Tuned, Tool-Calling Agent for Authorized Penetration Testing



Built on Qwen3.5-2B · QLoRA fine-tuned · Q6_K GGUF export

Hackathon Submission · August 2026

Pentest workflows still run on hand-typed commands

1

Slow handoffs

Translating an engagement plan into exact scanner/tool syntax costs time every step of the way.

2

Inconsistent scope discipline

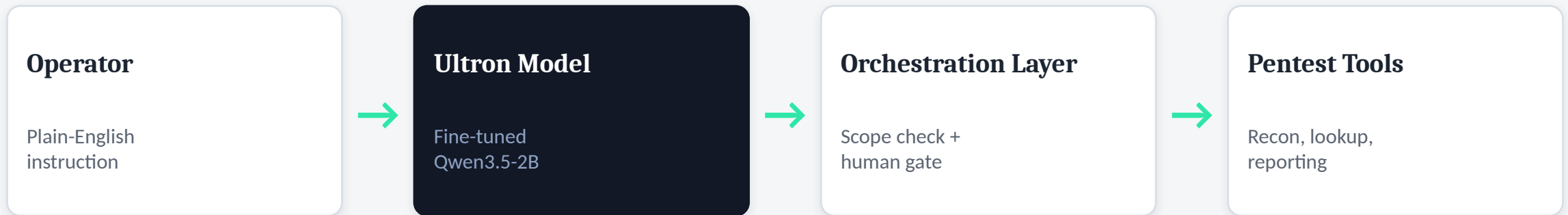
Manual workflows make it easy to drift outside an authorized target list under time pressure.

3

No structured trail

Ad hoc commands and notes make it hard to reconstruct exactly what was run, when, and why.

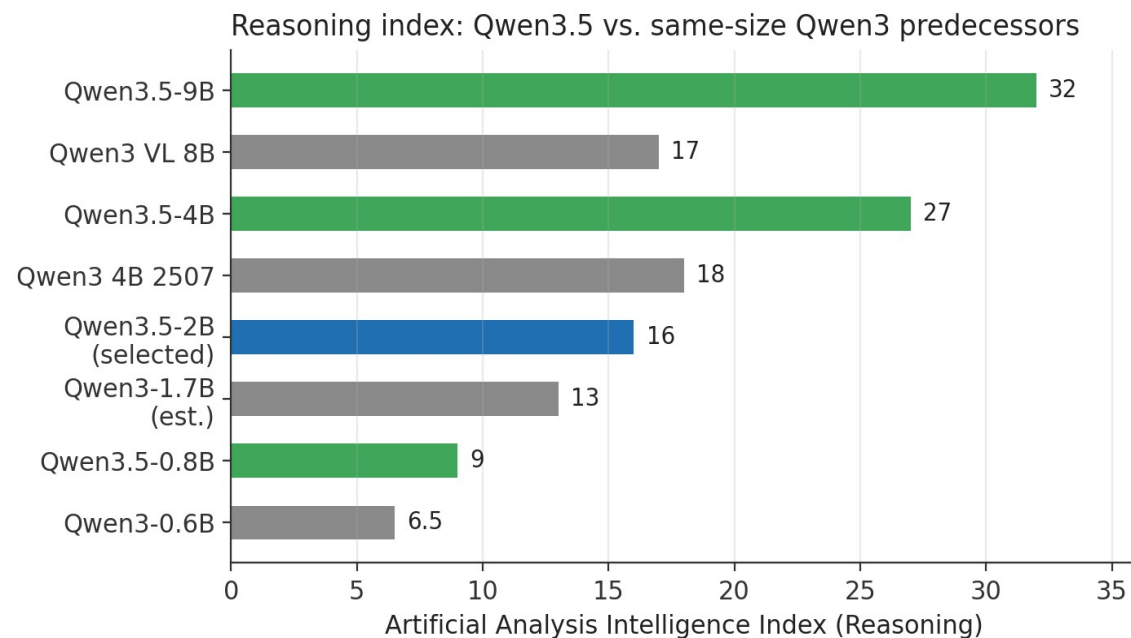
Ultron: instruction in, authorized tool call out



Every call from the model is validated against the engagement's authorized scope before anything executes — the model proposes, the orchestration layer decides.

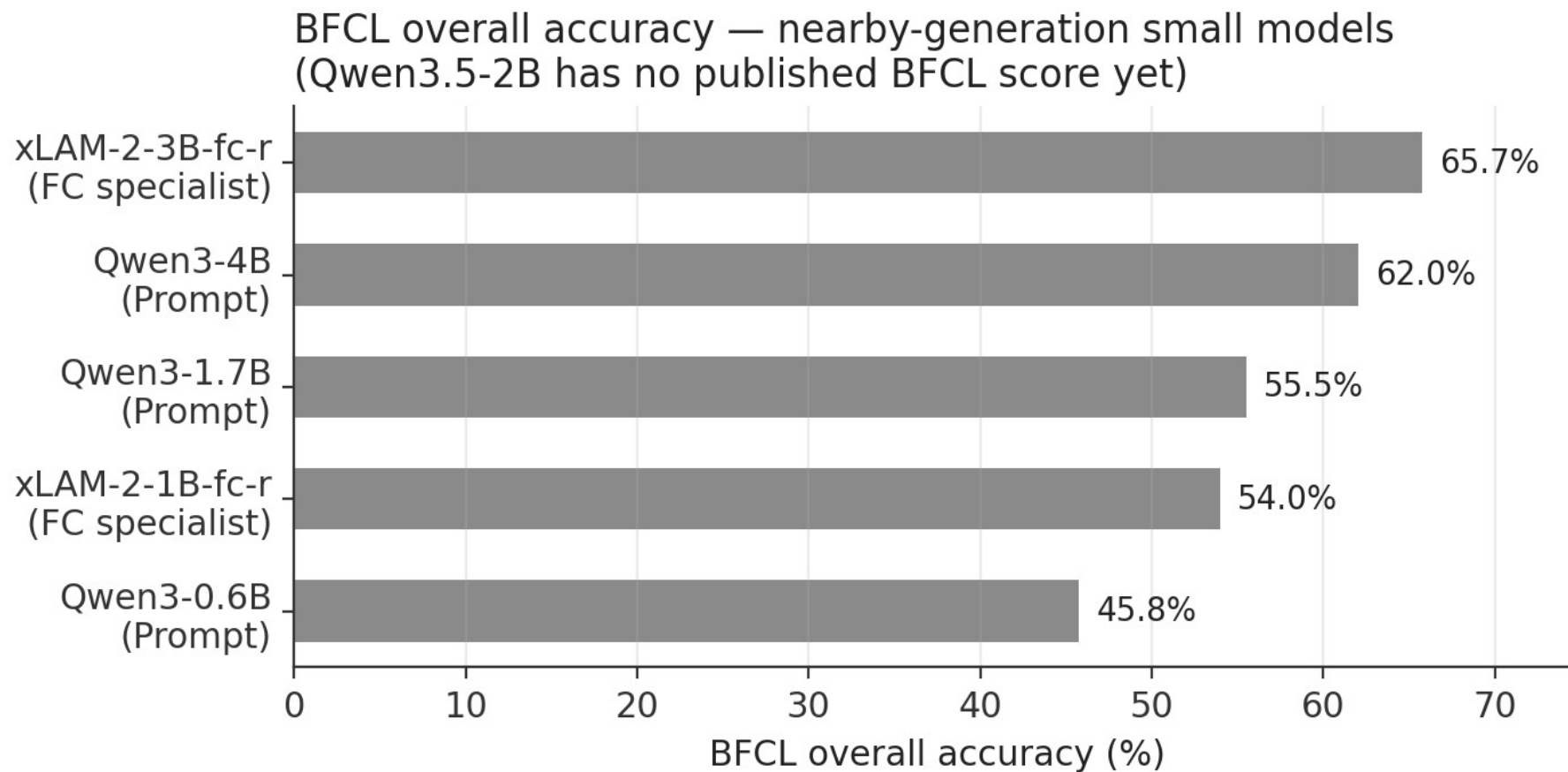
Why Qwen3.5-2B

- Native, OpenAI-compatible function calling
- Best reasoning-per-parameter at the 2B tier
- 262K native context for long tool + recon history
- Apache 2.0 — unrestricted commercial use
- Efficient hybrid architecture for continuous local inference



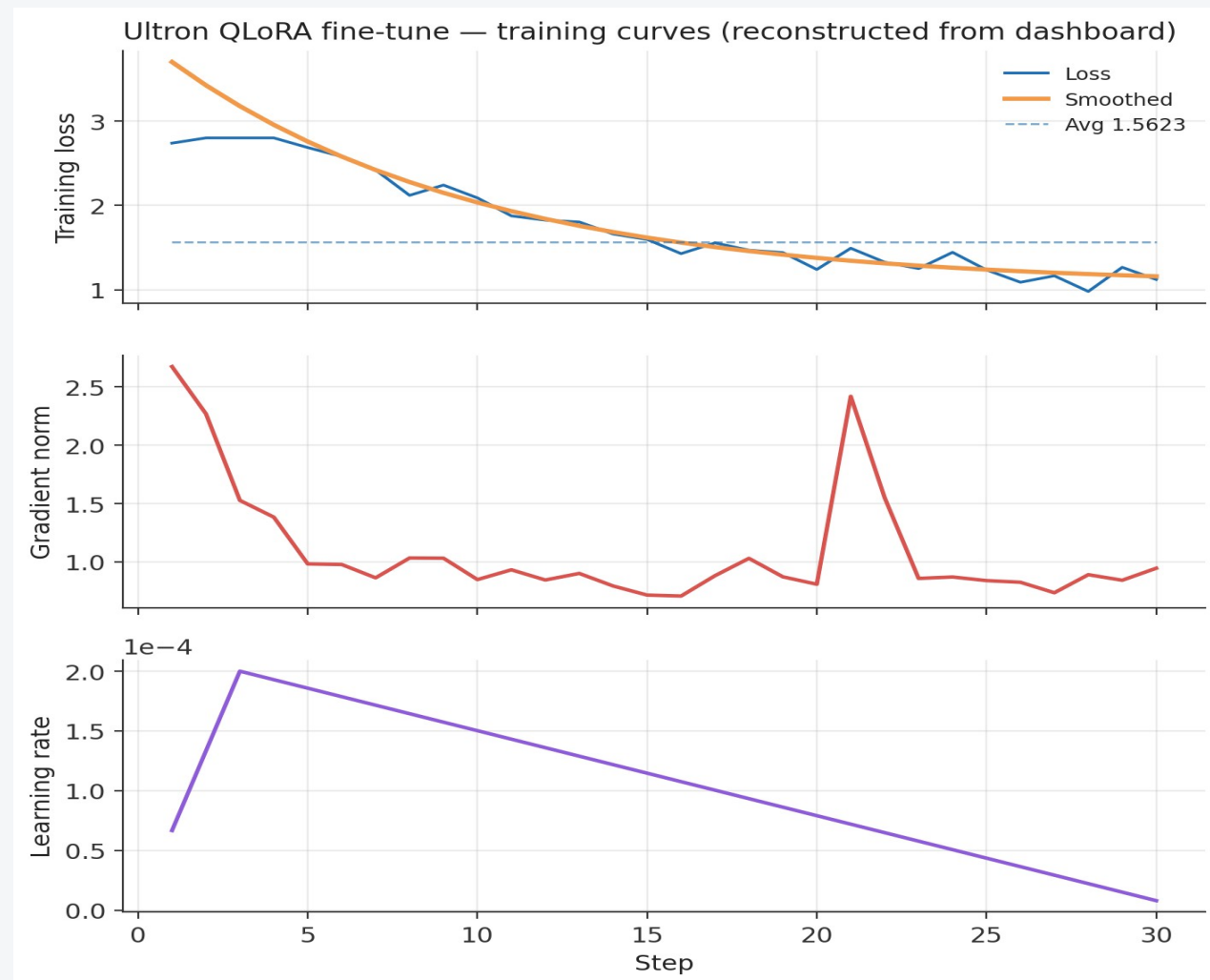
Artificial Analysis Intelligence Index — Qwen3.5-2B leads same-size Qwen3 predecessors.

How small tool-calling models compare



BFCL overall accuracy, single consistent benchmark run across nearby-generation small models (Qwen3.5-2B has no published BFCL score yet).

QLoRA fine-tuning converged in 30 steps

**1.2061**

Final training loss

30 / 30

Steps completed

40,580

Tokens processed

0.993

Final gradient norm

⚠ No held-out evaluation configured yet — these figures reflect training-set fit only.

Guardrails are designed in, not bolted on

Scope enforcement

No tool runs against a target that isn't registered under a signed authorization reference.

Human-in-the-loop

Medium/high-risk actions pause for explicit operator approval before execution.

Full audit trail

Every call, argument, and result is logged per engagement for review.

Untrusted tool output

Results returned by tools are treated as untrusted input, not as new instructions.

From hackathon build to field-ready

Now

Working QLoRA fine-tune, Q6_K GGUF export, scoped tool schema, demo-ready orchestration layer.

Next

Held-out evaluation + base-vs-fine-tuned comparison; prompt-injection test suite for tool outputs.

Later

Expand training data beyond the initial 40K-token run; independent review of scope-enforcement code.

Thank you

Ultron — fine-tuned tool calling, with authorization built into every step.